

Brain Dumpd :CLAD Prompt Log

Project: Brain Dumpd, a Snap SPECS AR Lens

Built with: CLAD + Claude Code driving Lens Studio via the Lens Studio MCP tools

Tooling used: Lens Studio MCP (VirtualScene, scene-graphql, asset-graphql, RecompileTypeScript, RunAndCollectLogs, PreviewPanel/screenshots, WorldQuery, IconSelector, GenerateFast3DAssets, KnowledgeBase), TypeScript for all Lens scripts, OpenAI (gpt-4o-mini) via the Remote Service Gateway.

How the AI-assisted workflow ran

Every feature followed the same loop: I described what I wanted → Claude proposed an approach (and asked focused questions when a decision was mine to make) → it wrote/edited the TypeScript, built the scene through the MCP tools, recompiled, ran the lens, read the runtime logs, and captured preview screenshots to verify before moving on. Data-layer logic (parsing, ring capacity, persistence) was covered with runnable tests. Design directions were chosen from side-by-side options rather than guessed.

Below is a representative, chronological log of the prompts I gave:

Phase 1 — Setup & design

- "Can you access my Lens Studio folder along with the CLAD MCP?"
- "I'm building a SPECS Lens called Brain Dumpd for a hackathon, Users can speak an unstructured brain dump, an LLM turns it into tasks with urgency, shown in my space. Produce a design document."
- "Install all the required packages."
- "Which LLM should we use?" → "Lock GPT." → "Overflow policy — option A."
- "Completion behaviour = manual; ASR path = voice ML; placement trigger = pinch to place; immediate for demo snappiness, confirmation as a toggle."

Phase 2 — Data layer (voice → tasks)

- "I'm done with Snapchat login + token. Implement the data layer only — strict JSON array of tasks, keep the LLM call in its own module."
- "Harden the parsing layer. Add a local fallback parser, then write and run tests."
- "Implement the ring/column capacity logic as pure functions. Write tests."
- "Update the design doc and wire TaskStore to RingCapacity."
- "Persist all tasks between sessions. Verify with a test."

Phase 3 — Design pivot (rings → sticky wall)

- "I don't like the rings — what can we redesign it to?"
- "A Calder-style mobile... what do you think of something like this?"
- "What I didn't like: urgency = distance, river of time, anti-overwhelm. I liked sticky notes on your real wall the most."

- "1. Columns. 2. I'm okay with the colours you suggested. Start building."

Phase 4 — Placement, persistence & interactions

- "Increase the size of the text; the sticky notes aren't attaching to any walls in preview — why?"
- "Wire the World Query wall placement now. Also add persistence module."
- "Can't see NEXT and LATER tags, and the re-anchor button isn't working. At the start the user should see a place-anchor placeholder."
- "No surface in view even when I'm clearly facing a wall."
- "Wire it to real data and build grab/peel interactions. The clear-cards and reattach-to-a-different-wall buttons are missing."
- "Continue with the buttons and grab/peel." / "Increase the size of the text, change the colour of the buttons."

Phase 5 — Voice states & mic button

- "Add listening and thinking states. While recording, show a pulsing indicator on the surface that responds to input volume. While waiting on the LLM, show a compact processing state in the same place. Also add a rounded button with a 2D mic that tap/pinch toggles recording."

Phase 6 — Crumple-to-delete (physics)

- "Is it possible that if I grab a card and throw it or drag it down, it crumples into a ball, drops on the floor, and is deleted?"
- "Build it with physics — and a crumple custom shader. Let's discuss."
 - (Chose: real generated crumpled-paper-ball mesh over a vertex shader.)
- "Can the crumpled paper ball be the same colour as the note that's been removed?"

Phase 7 — Button & toolbar redesign

- "Its perfect. Can we work on how the buttons look now?"
 - (Chose: rounded pill + icon + label.)
- "I want the mic button to be circular. Remove icons from Sweep wall and New wall. Keep icon only for mic (no text)."
- "Make the mic disc smaller and use a smoother circle."
- "The record indicator and LLM-working indicator need to be polished too."

Phase 8 — Headers, roaming mic & reminders

- "1. The backplates for NOW/NEXT/LATER don't look good — redesign them. 2. The mic should follow me into another room; when I speak elsewhere, notes are made in front of me then fly to the wall with persistent notes. 3. If a note mentions a time and lands in NOW, can I be reminded?"
 - (Chose: underline-tab headers; full roaming mic + fly-to-board; in-session reminder — Spectacles can't fire notifications while the Lens is closed.)

Phase 9 — Parsing bug + LLM debugging

- "The texts aren't parsing properly. Nothing goes into LATER, and 'I have a meeting with John tonight at 10:30' forms two different notes."
- "Yes, debug — I've regenerated the tokens now."

- (Root cause: an expired OpenAI token had been silently falling back to the local keyword parser; regenerating fixed it. Also hardened: normalize "10;30" → "10:30", stop splitting on ";", real LATER detection, tighter prompt.)

Phase 10 — Context-aware mic & motion polish

- "If I'm in the same room as the persistent wall, the mic sits on the wall like before; if I go to another room, it should come with me."
- "Decrease the size of the following mic button by 30%."
- "Make the notes flying to the wall a bit slower — they should first be generated in front of the user, then fly away."

Phase 11 — Final fixes, onboarding

- "A few fixes: 1. When the mic is on the wall and I speak, the note is generated twice (one on the wall and one animated from the mic) — the fly-in should only happen when the mic is floating and I'm away from the wall. 2. The 'time mentioned' alarm should last a bit longer and have a sound. 3. Add a start/onboarding screen explaining what does what, before we select the wall."
 - (Discovered a Lens constraint: a Lens can't mix licensed and non-licensed audio — since the mic uses a non-licensed Microphone track, the reminder sound had to be a non-licensed SFX, not a music-library track. Started with the "UI SFX Pack" notification sound, later swapped manually by the creator — see Phase 14.)

Phase 12 — Post-it styling & longer columns

- "1. I want the onboarding to look like a post-it note. 2. The wall-placement placeholder is currently a white plane — make it the onboarding post-it note instead. 3. Increase the column length of Now, Next and Later."
 - (Restyled the onboarding welcome as a tilted yellow sticky note with dark handwriting-style text; swapped the white placement placeholder for a matching yellow post-it that follows your gaze; raised the per-column capacity and dropped the toolbar down so the columns run longer.)

Phase 13 — Mic visibility, placeholder hint & note spacing

- "1. The mic should be visible only after the wall placement is done. 2. When the user changes the wall, only a blank post-it is visible — can we have 'place on a flat surface' text appear 3. The first note on the board overlaps the header underline in all three columns."
 - (Mic HUD now scales to zero and hides during onboarding + placement, appearing only once the wall is locked; added a "Place on a flat surface" label to the placeholder post-it; nudged the top of each column down so the first note clears the header underline.)

Phase 14 — Alarm loop & mic size

- "1. The alert sound should play on loop for the amount of time the note is pulsing. 2. Make the floating mic's size a little smaller. Note: I've changed the sound file manually."
 - (Reminder chime now loops (play) for the full 8s bounce and is stopped/faded when it ends, instead of playing once; shrank the roaming mic's follow scale from 0.7 to 0.55. The reminder audio clip itself was swapped manually by the creator — the exact sound is whatever is wired into the StickyWall `reminderSound` input.)

This log is a faithful, condensed record of the prompts and the build-verify workflow used with CLAD to create Brain Dumpd.